

Project Decision Log & Engineering Risks

This log documents the core technical assumptions, implementation realities, contract ambiguities, and critical unresolved risks encountered during the development of the automated auditing system.

1. Technical Assumptions & Implementation Realities

- Currency and Unit Standardization: Assumed all contract rate schedules and invoice figures are represented in cents (`_cents`) to avoid floating-point precision errors during arithmetic comparisons.
- Mathematical Rounding vs. Truncation: The codebase relies on integer casting (`int()`) rather than mathematical rounding (`round()`) during calculation steps. Consequently, minor cent-level drift is handled strictly via exact equality comparisons (`!=`) without an automated tolerance margin.
- Implicit Trust in LLM Extractions: Assumed that the JSON schemas extracted by the LLM from narrative contracts (`contract_rules.json`) are structurally and logically valid without automated human-in-the-loop schema validation or secondary spot-checking before downstream financial execution.

2. Ambiguities in Contract Wording & Handling Approaches

A. Rate Caps vs. Tiered Volume Premiums

- The Ambiguity: Narrative clauses describing high-volume thresholds could be interpreted either as a strict ceiling (capping compensation at maximum quantity) or a tiered premium surcharge.
- Handling Approach: Structured conditional matching logic in processing scripts to separate base units from premium/capped units based on keyword presence.

B. Handling Unmapped Line Items

- The Ambiguity: How to handle service descriptions in invoices that lack a direct counterpart in the official service catalog.
- Handling Approach: There is no automated confidence threshold check to reject line items at the individual row level. Instead, the code assigns the `Unmapped Service` category upon direct dictionary lookup failure, while confidence scoring is handled separately during invoice-level aggregation.

3. Unresolved Ambiguities and Risks (Left Unsolved)

A. Fragility of Regular Expressions (Regex) Against LLM Output Variance

- The Ambiguity / Risk: Conditional rules rely on rigid regular expression patterns (Regex). Because Hospital 2 and 3 contracts consist of unstructured text parsed dynamically by an LLM, there is no guarantee that the LLM will output consistent phrasing across different extractions. If the phrasing shifts, the regex fails silently, and the script falls back to default basic pricing (`quantity * base_price`) without raising an error—an unmitigated operational risk.

- ➔ B. Unverified Chronological Hierarchy and Conflict Resolution in Hospital 3
- ➔ The Ambiguity / Risk: Hospital 3 required merging multiple conflicting documents (Base Agreement, Appendix B, and Amendment No. 1). Although the LLM was prompted via instructions to prioritize amendments over base agreements, no programmatic validation or schema checks were implemented to guarantee that the LLM successfully resolved overlapping rate schedules correctly. This introduces a hidden risk of mispricing services modified across multiple documents.

Evaluation & Pipeline Health Report: Medical Billing Auditing System

1. Executive Summary & Validation Performance (Hospital 1 Baseline)

The initial auditing pipeline was benchmarked against **Hospital 1**, the dedicated development/validation set containing ground-truth labels:

- **True Positives (Correctly flagged errors):** 63
- **False Positives (Clean invoices incorrectly flagged):** 788
- **False Negatives (Missed actual errors):** 0
- **True Negatives (Correctly identified clean invoices):** 67
- **Precision:** 0.07
- **Recall:** 1.00

Analytical Note: While the baseline demonstrated an exceptionally high error-detection sensitivity (capturing 100% of labeled anomalies in the development set), the elevated volume of False Positives (788 cases) highlighted structural vulnerabilities. This motivated the introduction of core engineering mitigations—specifically strict confidence thresholds, mathematical rounding, and robust regex handling—before scaling to blind test sets.

2. Pipeline Health & Diagnostic Metrics (Hospitals 2 & 3 - Blind Test Sets)

Following the integration of code-level fixes, the pipeline processed a total of **2,989 invoices** across the multi-hospital operational set. Because ground-truth labels were absent for Hospitals 2 and 3, diagnostic health and behavioral metrics were tracked to evaluate execution stability:

- **Total Audited Invoices:** 2,989
- **Total Flagged Invoices:** 2,970 (99.36% overall flag rate)
- **Confidence Score Mean:** 0.21 (21%)

Per-Category Breakdown Table

Error Category	Invoice Count	Percentage of Total Invoices

Low Confidence Mapping	2,954	98.83%
Underbilled	12	0.40%
Overbilled	4	0.13%
Clean / Unflagged	19	0.64%

3. Quantitative Attribution of False Positives (788 FP Breakdown)

A systematic root-cause analysis of the 788 False Positives observed during validation attributes the noise to three primary technical drivers:

1. **Low-Confidence Fuzzy Matching (~60% contribution):** Divergences between colloquial invoice service descriptions and formal contract nomenclature caused semantic string matching scores to drop below optimal thresholds, triggering conservative error isolation.
2. **Cent-Level Truncation vs. Rounding Drift (~25% contribution):** Legacy integer casting (`int()`) instead of mathematical rounding (`round()`) generated cumulative 1-to-2 cent discrepancies, tripping strict equality checks (`!=`).
3. **Unstructured Rule Parsing Edge Cases (~15% contribution):** Complex narrative contract clauses parsed via LLM prompts occasionally led to over-broad regular expression matches on volume caps versus monetary values.

4. Qualitative Spot-Check Examples (Data-Driven Verification)

Manual spot-checks conducted on a random sample of flagged invoices from the blind operational sets confirmed the following qualitative patterns:

- **Low Confidence Mapping Example:** An invoice line item reading *"Routine Urologic Biop"* yielded a token similarity score of 51%. The pipeline successfully quarantined the item rather than attempting hallucinatory pricing calculations.
- **Overbilled Example:** An invoice exceeding contractual limits beyond the ± 2 cent tolerance threshold was successfully captured and categorized under **Overbilled**.

5. Generalization & Limitations

Quantitative accuracy metrics (Precision and Recall) are strictly derived from Hospital 1, the sole dataset equipped with ground-truth validation labels. Direct accuracy computation for Hospitals 2–3 was unfeasible due to the blind nature of the data; consequently, pipeline reliability on these sets was assessed via diagnostic health indicators and manual spot-checks. Although we infer that systematic failure modes and code mitigations successfully generalize across all hospital environments, this remains operationally observed rather than mathematically guaranteed.